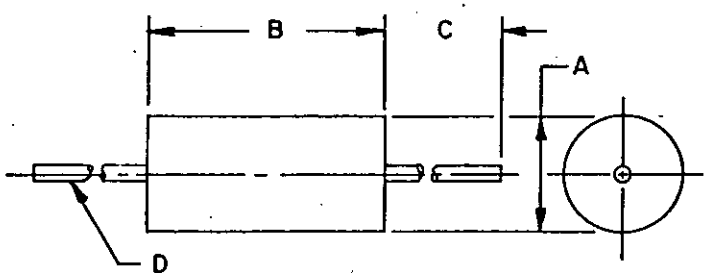
MILITARY SPECIFICATION SHEET

COILS, FIXED, RADIOFREQUENCY, MOLDED, SUB-MINIATURE (POWDERED IRON CORE), ESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-39010.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.157 (3.99)	.219 (5.56)
B	.407(10.34)	.469(11.91)
C	1.250(31.75)	1.626(41.30)
D	.023 (.58)	.027 (.69)

NOTE: Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Material: Powdered iron core.

Weight: 0.950 gram maximum.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage: Method 301 of MIL-STD-202 test voltage: 1,000 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202 (70,000 feet), test voltage 200 volts rms.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Inductance tolerance: See table I.

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Q values: See table I.

Self-resonant frequency: See table I.

DC resistance: See table I.

Temperature rise: 15°C.

Terminal pull: 5 pounds.

Part number: M39010/05 (dash number from table I).

Revision letters are not used to denote changes due to the extensiveness of the changes.

TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/
A3R0**	3.00	5	30	7.9	70	0.140	945
A3R3**	3.30	5,10	30	7.9	70	0.140	945
A3R6**	3.60	5	30	7.9	65	0.155	870
A3R9**	3.90	5,10	30	7.9	65	0.155	870
A4R3**	4.30	5	30	7.9	60	0.210	745
A4R7**	4.70	5,10	30	7.9	60	0.210	745
A5R1**	5.10	5	30	7.9	50	0.280	645
A5R6**	5.60	5,10	30	7.9	50	0.280	645
A6R2**	6.20	5	30	7.9	50	0.375	560
A6R8**	6.80	5,10	30	7.9	50	0.375	560
A7R5**	7.50	5	30	7.9	48	0.440	520
A8R2**	8.20	5,10	30	7.9	48	0.440	520
A9R1**	9.10	5	30	7.9	42	0.605	440
A100**	10.0	5,10	30	7.9	42	0.605	440
A110**	11.0	5	30	2.5	36	1.05	335
A120**	12.0	5,10	50	2.5	36	1.05	335
A130**	13.0	5	50	2.5	30	1.20	310
A150**	15.0	5,10	55	2.5	30	1.20	310
A160**	16.0	5	55	2.5	30	1.95	245
A180**	18.0	5,10	60	2.5	30	1.95	245
A200**	20.0	5	60	2.5	24	2.20	230

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/
A220**	22.0	5,10	60	2.5	24	2.20	230
A240**	24.0	5	60	2.5	22	2.75	205
A270**	27.0	5,10	55	2.5	22	2.75	205

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the failure rate symbol (e.g., M, P, R, S) and will be added to the end of the dash number.
2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.

TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from initial measurement		Allowable % from specified minimum value in electrical characteristics (initial) table	
	Inductance	DC resistance	Self-resonant frequency	Q
	Percent			
Qualification inspection				
Group II	±2	---	---	-10
Group IV	±5	±(3%+.001 ohm)	1/	-10
Group V	±5	±(2%+.001 ohm)	1/	-15
Quality conformance inspection				
Group C				
Subgroup 1	±5	±(2%+.001 ohm)	1/	-15
Subgroup 3	±2	---	---	-10
Subgroup 4	±5	±(3%+.001 ohm)	1/	-10

1/ The self-resonant frequency shall be not less than the value specified in table I.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, CG, MC, SH
Air Force - 19

Agent:
DLA-ES

Preparing activity:
Army - ER

(Project 5950-0603-5)

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